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Ministry of Education
King Faisal University
College of Computer Sciences & Information Technology
Department of Computer Sciences

King Faisal University

Department of Computer Sciences

Project Report

Playfair Cipher Encryption and Decryption System

Computer Security
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CS 320

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Project Marking Guide (Programming Project)

	Assignment Component	Max. Marks	Marks Obtained
REPORT	Format (Table of Contents, References, formatting and font, etc....)	10	
	English language (grammar and spelling)	10	
	Introduction	15	
	References	10	
	Contents Algorithms, and Discussions / Coding	30	
Demonstration	Questions and answers	15	
	Interface Design & Usability	15	
	Testing (Various Input Conditions)	15	
	Presentation skills	15	
	Understanding / Explanation of the code	15	
	Total Marks	150	
	Grade for this Project	15	
	Plagiarism report more than 10% to 25% Plagiarism report more than 25% to 40% Plagiarism report more than 40%	- 5 -10 -20	

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Introduction

Cryptography is a fundamental concept in computer security that focuses on protecting information from unauthorized access. It involves transforming readable data, known as plaintext, into an unreadable format called ciphertext. This ensures confidentiality, integrity, and secure communication between users.

With the increasing reliance on digital systems, the importance of cryptography has significantly grown. It is widely used in applications such as online banking, secure messaging, authentication systems, and data protection. Without proper encryption techniques, sensitive information can easily be intercepted or manipulated by attackers.

One of the classical encryption techniques is the Playfair Cipher. Unlike simple substitution ciphers that encrypt one letter at a time, the Playfair Cipher encrypts pairs of letters (digraphs), making it more secure and resistant to frequency analysis attacks. It was invented in 1854 and became popular for manual encryption due to its simplicity and effectiveness.

This project focuses on designing and implementing a Playfair Cipher system that can perform both encryption and decryption. The system allows users to input a keyword to generate a key matrix and then encrypt or decrypt messages based on that matrix.

System Description

The developed system is designed to provide a simple and user-friendly way to perform encryption and decryption using the Playfair Cipher technique.

Inputs

- Key (keyword provided by the user)
- Plaintext (for encryption)
- Ciphertext (for decryption)

Outputs

- Encrypted text (ciphertext)
- Decrypted text (plaintext)

The system first generates a 5×5 matrix using the provided keyword. This matrix is used as the basis for both encryption and decryption processes.

A graphical user interface (GUI) is included to improve usability. The interface contains:

- Input fields for key and text
- Buttons for encryption and decryption
- Output display area for results

The system ensures proper handling of input text by:

- Converting all letters to uppercase
- Removing spaces
- Replacing the letter “J” with “I”
- Splitting the text into pairs of letters
- Adding filler characters (such as “X”) when needed

This design makes the system efficient, easy to use, and aligned with classical cryptographic principles.

Algorithm

The Playfair Cipher algorithm consists of two main parts: key matrix generation and the encryption/decryption process.

1. Key Matrix Generation

A 5×5 matrix is created using the keyword provided by the user.

Steps:

1. Remove duplicate letters from the keyword.
2. Convert all letters to uppercase.
3. Replace the letter “J” with “I”.
4. Fill the matrix row by row using the keyword.
5. Fill the remaining spaces with unused letters of the alphabet (excluding J).

Example:

Keyword: SECURITY

Matrix:

S E C U R

I T Y A B

D F G H K

L M N O P

Q V W X Z

2. Plaintext Preparation

Before encryption, the plaintext must be processed:

- Convert to uppercase
- Remove spaces
- Replace J with I
- Split into pairs

Example:

HELLO → HE LL O → HE LX LO

3. Encryption Rules

Each pair of letters is encrypted using the following rules:

Rule 1: Same Row

If both letters are in the same row:

→ Replace each letter with the one to its right

Rule 2: Same Column

If both letters are in the same column:

→ Replace each letter with the one below it

Rule 3: Rectangle Rule

If letters form a rectangle:

→ Replace each letter with the one in the same row but different column

4. Decryption Rules

Decryption is the reverse process:

- Same row → move left
- Same column → move up
- Rectangle → same rule

Implementation

The Playfair Cipher system was implemented using Java with a Graphical User Interface (GUI) built using the Swing library. The program is divided into three main parts:

1. PlayfairMainGUI

This class handles the user interface, including input fields for the key and plaintext, buttons for encryption and decryption, and an output area to display results. It also manages user interactions and validates input.

2. PlayfairPart1

This class is responsible for preparing the plaintext and generating the 5×5 key matrix. The prepareText() function processes the input by removing spaces, converting letters to uppercase, replacing 'J' with 'I', and splitting the text into valid pairs. The generateMatrix() function creates the matrix using the key while ensuring no duplicate letters.

3. PlayfairPart2

This class performs encryption and decryption. The encrypt() and decrypt() functions apply the Playfair Cipher rules. The processPair() method implements the three main rules: same row, same column, and rectangle substitution.

Overall, the system follows the standard Playfair Cipher algorithm and ensures correct transformation of plaintext into ciphertext and vice versa.

Interface Design

Initial Interface

This figure shows the main interface of the Playfair Cipher system. The user can enter the encryption key and plaintext in the provided fields. The interface includes two buttons for encryption and decryption, along with an output area. At this stage, no operation has been performed yet.

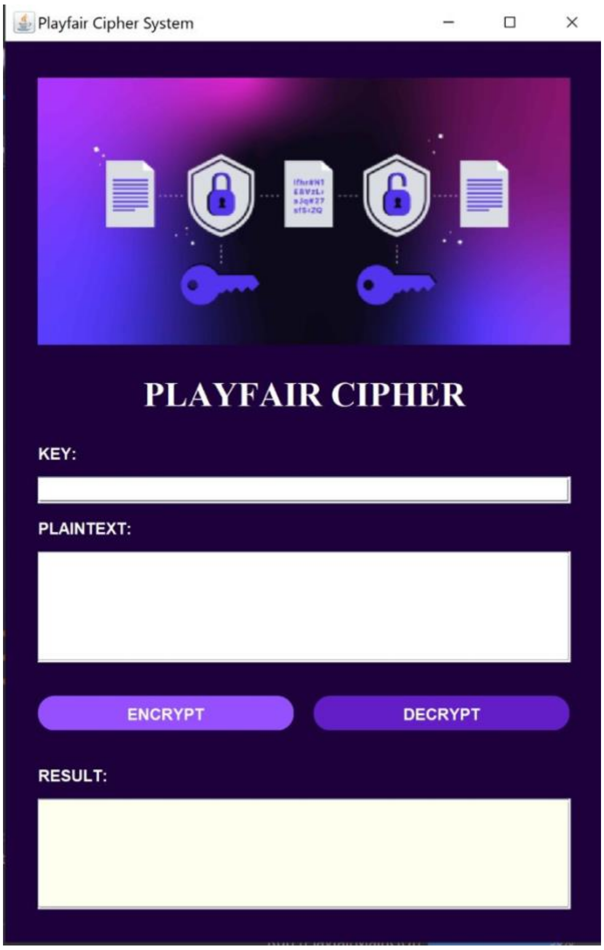

Figure 1: Initial Interface

Encryption Process

This figure demonstrates the encryption process. After entering the key and plaintext, the user clicks the "Encrypt" button. The system processes the input using the Playfair Cipher algorithm and displays the resulting ciphertext in the output area.

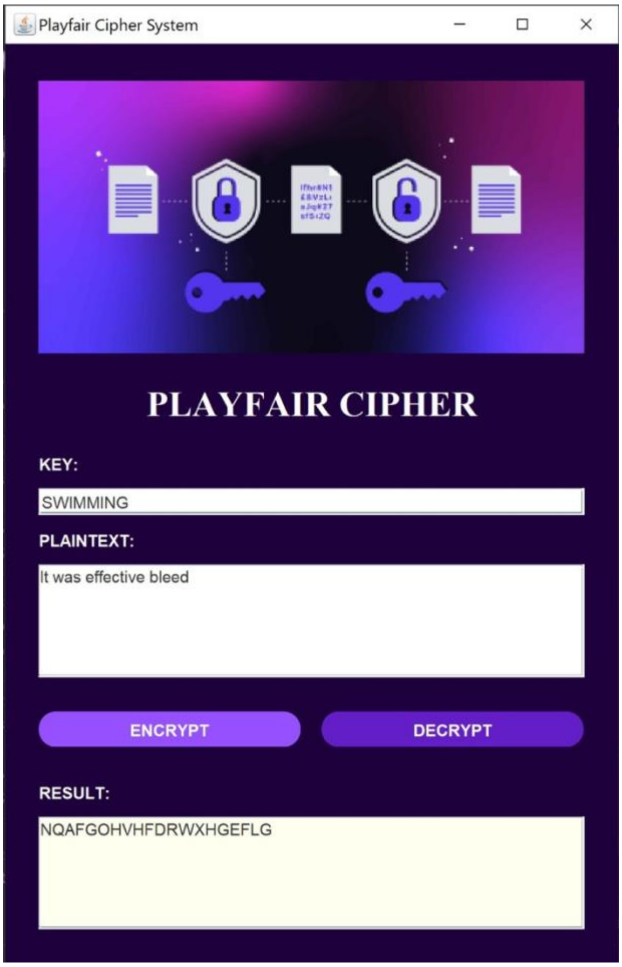

Figure 2: Encryption Process

Decryption Process

This figure shows the decryption process. The user enters the key and ciphertext, then clicks the "Decrypt" button. The system applies the decryption rules and successfully retrieves the original plaintext.

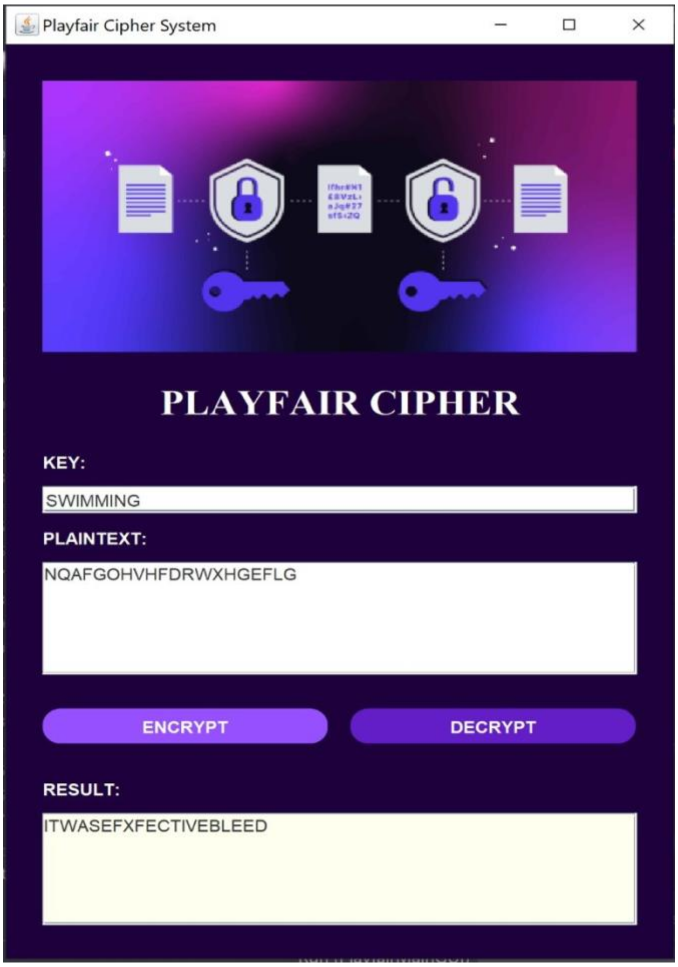

Figure 3: Decryption Process

Testing

The system was tested using the following example:

Key: SWIMMING
Plaintext: IT WAS EFFECTIVE BLEED

The system processed the input and produced the following ciphertext:

Ciphertext: NQAFGOHVHFDRWXHGEFLG

The decryption process was also tested using the same key, and the system successfully retrieved the original plaintext.

Additional tests were performed to ensure robustness, including:
- Repeated letters (e.g., EFFECTIVE)
- Lowercase input (e.g., it was bleed)
- Empty input validation

All test cases produced correct and expected results.
This confirms that the implemented system functions correctly according to the Playfair Cipher rules.

Discussion

The Playfair Cipher offers several advantages compared to simpler classical encryption techniques. By encrypting pairs of letters instead of single characters, it significantly reduces the effectiveness of frequency analysis attacks.

Strengths

- More secure than monoalphabetic substitution ciphers
- Simple to implement

- Efficient for manual encryption
- Provides better confusion in ciphertext

Weaknesses

- Still vulnerable to modern cryptanalysis
- Limited key space compared to modern algorithms
- Requires preprocessing of text (handling repeated letters and spacing)
- Not suitable for large-scale secure systems

Despite these limitations, the Playfair Cipher remains an important educational tool. It helps students understand the fundamentals of encryption, key generation, and algorithm design.

In this project, implementing the Playfair Cipher provided practical experience in:

- Handling text processing
- Designing encryption algorithms
- Building user interfaces
- Understanding classical cryptographic systems

Questions and Answers

Q1: What is the main advantage of the Playfair Cipher?
A: It encrypts pairs of letters instead of single letters, making it more secure than simple substitution ciphers.

Q2: Why is the letter 'J' replaced with 'I'?
A: Because the Playfair Cipher uses a 5×5 matrix, which can only contain 25 letters.

Q3: What happens when two identical letters appear in a pair?
A: The system inserts the letter 'X' between them to avoid duplication.

Conclusion

In this project, a Playfair Cipher encryption and decryption system was successfully designed and implemented. The system allows users to securely transform plaintext into ciphertext and recover it using the same key.

The project demonstrates the importance of classical cryptography and its role in understanding modern security techniques. While the Playfair Cipher is not suitable for modern secure communication, it provides a strong foundation for learning encryption concepts.

The inclusion of a graphical user interface enhances usability and makes the system more interactive. Overall, the project achieved its objectives and provided valuable insights into cryptographic systems.

Future improvements may include:

- Supporting additional encryption techniques
- Enhancing the user interface
- Integrating modern encryption algorithms

References

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